The Jacobian lens across medical fine-tuning

Phase 1 — first real trajectory: Gemma-3-1b-it

A training-dynamics / interpretability study. We re-fit the average input→output Jacobian lens (J_ℓ) at every checkpoint and track how the internal read-out and its faithfulness to the model's own output evolve as medical SFT proceeds.

base + 14 log-spaced checkpoints · fit on 1000×128 wikitext (110,741 source positions) · 193-prompt probe set (50 general / 50 medical / 93 multihop) · 2026-07-24

What was measured

Model / run

- Gemma-3-1b-it, full fine-tune on `mix_default`
- lr 1×10^{-5}, 2 epochs, seed 42, 4882 steps
- 26 layers, $d_{\text{model}}=1152$
- log-spaced checkpoints $\{1,2,4,\ldots,4096,4882\}$; $t=0$ is the untouched base

Per checkpoint (re-fit J_ℓ each time)

- **Drift** of J_ℓ vs the $t=0$ lens
- **Faithfulness**: KL(model ‖ lens) and top-1 agreement, per source layer

The lens is a function of the weights, so it is re-fit on a *frozen* generic corpus at every checkpoint.

Faithfulness is read on a *frozen* 193-prompt set: a **general** control, a **medical** arm (50 each, hand-authored cloze), and jlens's own **multihop** set (93, a curated reference). The study watches medical concordance move *relative to* the general control.

Compute: ~9.6 h train + 14 h probe on 1×A100.

The three metrics, defined

Let J_ℓ be the fitted Jacobian at source layer ℓ, J_ℓ^0 its $t{=}0$ (base) value, and at a probe position let p be the model's next-token distribution and q_ℓ the lens read-out transported from layer ℓ.

Drift (per layer, then mean)

$$r_\ell = \frac{\|J_\ell - J_\ell^0\|_F}{\|J_\ell^0\|_F}, \quad c_\ell = \frac{\langle J_\ell, J_\ell^0\rangle}{\|J_\ell\|\,\|J_\ell^0\|}$$

How far the lens has moved from the base. r rises from 0, c falls from 1.

Faithfulness KL

$$\mathrm{KL}(p\,\|\,q_\ell) = \sum_v p_v \log \frac{p_v}{q_{\ell,v}}$$

Divergence between what the model says and what the lens reads out at layer ℓ. Lower $=$ more faithful; sensitive at small probe sets.

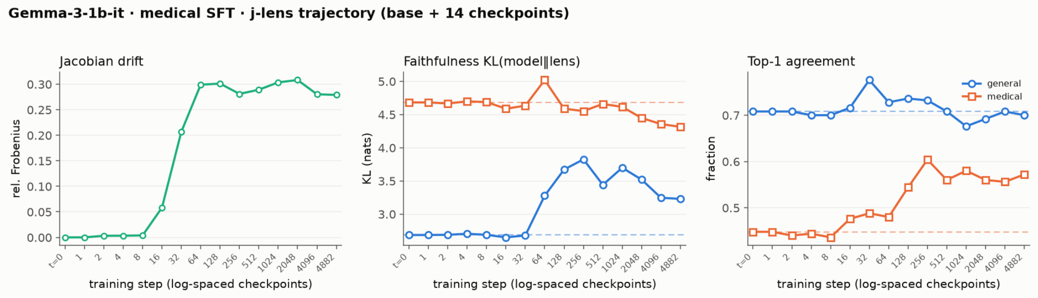
Top-1 agreement

$$a_\ell = \mathbf{1}[\hat{\imath}_\ell = \hat{\imath}]$$

$\hat{\imath}_\ell = \arg\max q_\ell$, $\hat{\imath} = \arg\max p$: does the lens's top token match the model's? Coarser than KL, so noisier.

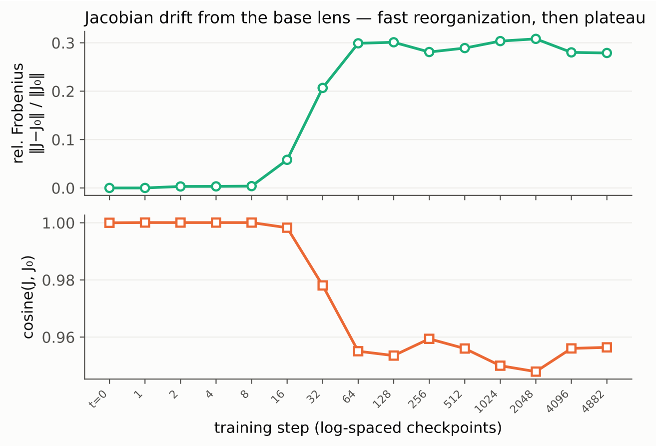
What the plots show. Drift panels: $\bar{r}, \bar{c}$ meaned over all source layers. KL / top-1 panels: $\overline{\mathrm{KL}}$ and $\bar{a}$ over the *final* $\lceil n_{\text{layers}}/5\rceil$ layers (where the read-out has resolved), averaged over the probe prompts and split by **general** / **medical** arm. All read at the last token position.

The trajectory at a glance



Drift ramps then plateaus; the **medical**–**general** KL gap *narrows* over training (**medical** improves, **general** degrades); **medical** top-1 rises toward the control. Dashed lines mark the $t=0$ base.

Drift: fast reorganization, then plateau

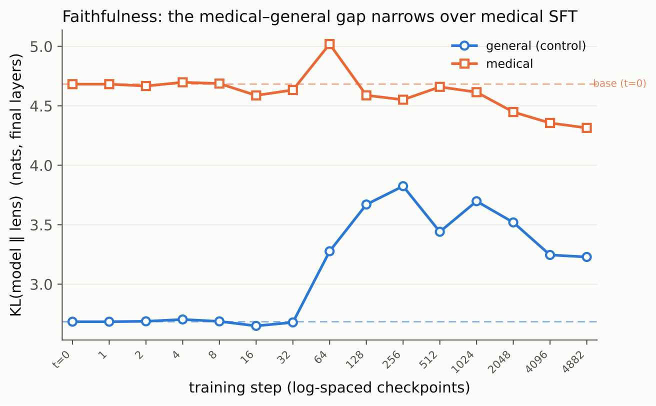


J_ℓ barely moves for the first $\sim$8 steps, reorganizes sharply over steps 16–64, and **plateaus at rel. Frobenius** ≈ 0.30 (cosine ≈ 0.95) for the remaining $\sim$4800 steps.

The representation's input$\rightarrow$output coupling settles early; later training is behavioral refinement, not further lens movement.

Caveat: `checkpoint-1` is bit-identical to $t{=}0$ (drift exactly 0, verified $\max|\Delta|{=}0$) — the warmup schedule applies lr$=0$ on step 1. Now dropped for future runs; still present here since this trajectory predates the fix.

Faithfulness: the medical–general gap narrows

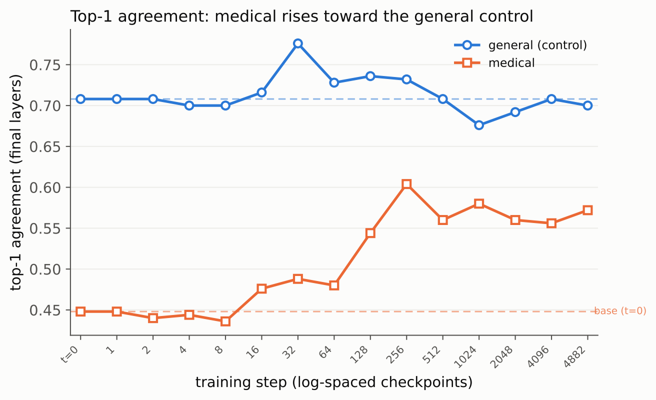


At $t=0$ the lens is far less faithful on medical (**medical** KL 4.68 vs **general** 2.68 — a **2.0-nat gap**). Over medical SFT that gap **halves to $\sim$1.1 nat**, from *both* sides:

general KL **degrades** $2.68 \to 3.23$ $(+0.54)$; **medical** KL **improves** $4.68 \to 4.31$ (-0.37). A domain-specific effect — exactly the differential the control is there to isolate.

Clean at 50/arm where it was buried at 15. Absolute KL is not comparable to earlier smaller-set runs (different prompts); per-prompt CIs not yet computed.

Top-1 agreement corroborates the KL story

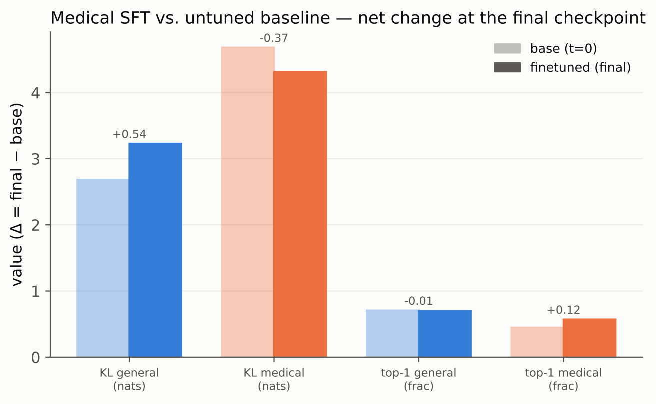


At 50/arm the coarser metric now agrees with KL: **medical** top-1 **rises** $0.45 \to 0.57$ toward the **general** control (flat ≈ 0.70). The lens's top token matches the model's more often on medical after SFT.

Still coarser and noisier than KL — **KL remains the workhorse** — but expanding the probe set from 15 to 50/arm pulled a real trend out of what was pure noise before.

multihop (reference arm) stays flat ≈ 0.5 — the effect is medical-specific, not a global shift.

Vs. the untuned baseline: a directional, domain-specific change



Endpoint summary — base ($t{=}0$) vs. the final checkpoint, the change medical SFT actually produced. (Every trajectory point is read against this dashed baseline.)

The two arms move in **opposite** directions: **medical** KL -0.37 and **medical** top-1 $+0.12$ (more faithful on medical), while **general** KL $+0.54$ and top-1 ≈flat (the control drifts the other way).

Not a global shift in concordance — a **medical-specific** realignment of the lens, on top of

Findings & next steps

Findings

- J_ℓ reorganizes in the first $\sim$64 steps, then plateaus — the largest single effect (drift $0 \to 0.28$).
- Medical SFT produces a **domain-specific faithfulness change**: the **medical**–**general** KL gap halves (**medical** -0.37, **general** $+0.54$) with **medical** top-1 $+0.12$, while the **multihop** reference stays flat.
- Expanding 15→50/arm turned that from noise into a clean, KL+top-1 trend.

Done since the first pass

- Probe set expanded to 193 (frozen); probe made **resumable** (GPU-verified); `checkpoint-1`≡base fixed for future runs.

Next

- Per-prompt CIs on the gap (is the narrowing significant, not just clean?).
- Pair with the behavioral trajectory (MedQA / MedMCQA / PubMedQA).
- Second real trajectory: 1b-*pt* (the pt-vs-it starting-point axis).